

| Table 50.1 Types of Skeletal Muscle Fibers |                     |                     |                    |
|--------------------------------------------|---------------------|---------------------|--------------------|
|                                            | Slow-Twitch         | Fast-Twitch         |                    |
|                                            | Oxidative           | Oxidative           | Glycolytic         |
| Contraction speed                          | Slow                | Fast                | Fast               |
| Major ATP source                           | Aerobic respiration | Aerobic respiration | Glycolysis         |
| Rate of fatigue                            | Slow                | Intermediate        | Fast               |
| Mitochondria                               | Many                | Many                | Few                |
| Myoglobin content                          | High (red muscle)   | High (red muscle)   | Low (white muscle) |

as their primary source of ATP and fatigue more readily than oxidative fibers. These two fiber types are readily apparent in the muscle of poultry and fish: The dark meat (red muscle) is made up of oxidative fibers rich in myoglobin, and the light meat (white muscle) is composed of glycolytic fibers.

**Fast-Twitch and Slow-Twitch Fibers** Muscle fibers vary in the speed with which they contract: **Fast-twitch fibers** develop tension two to three times faster than **slow-twitch fibers**. Fast fibers enable brief, rapid, powerful contractions. Compared with a fast fiber, a slow fiber has less sarcoplasmic reticulum and pumps  $\text{Ca}^{2+}$  more slowly. Because  $\text{Ca}^{2+}$  remains in the cytosol longer, a muscle twitch in a slow fiber lasts about five times as long as one in a fast fiber.

The difference in contraction speed between slow-twitch and fast-twitch fibers mainly reflects the rate at which their myosin heads hydrolyze ATP. However, there isn't a one-to-one relationship between contraction speed and ATP source. Whereas all slow-twitch fibers are oxidative, fast-twitch fibers can be either glycolytic or oxidative.

Most human skeletal muscles contain both fast-twitch and slow-twitch fibers, although the muscles of the eye and hand are exclusively fast-twitch. In a muscle that has a mixture of fast and slow fibers, the relative proportions of each are genetically determined. However, if such a muscle is used repeatedly for activities requiring high endurance, some fast glycolytic fibers can develop into fast oxidative fibers. Because fast oxidative fibers fatigue more slowly than fast glycolytic fibers, the result will be a muscle that is more resistant to fatigue.

Some vertebrates have skeletal muscle fibers that twitch at rates far faster than any human muscle. For example, superfast muscles produce a rattlesnake's rattle and a dove's coo. Even faster are the muscles surrounding the gas-filled swim bladder of the male oyster toadfish (**Figure 50.34**). In producing its "boat whistle" mating call, the toadfish can contract and relax these muscles more than 200 times per second!

▼ **Figure 50.34 Specialization of skeletal muscles.** The male oyster toadfish (*Opsanus tau*) uses superfast muscles to produce its mating call.



## Other Types of Muscle

Although all muscles share the same fundamental mechanism of contraction—actin and myosin filaments sliding past each other—there are many different types of muscle. Vertebrates, for example, have cardiac muscle and smooth muscle in addition to skeletal muscle (see Figure 40.5).

Vertebrate **cardiac muscle** is found only in the heart and, like skeletal muscle, is striated. Unlike the cells of skeletal muscle fibers, some cardiac muscle cells can initiate rhythmic depolarization and contraction without nervous system input. Normally, however, cells in one part of the heart act as a pacemaker to initiate contraction. Signals from the pacemaker spread throughout the heart because specialized regions called *intercalated disks* electrically couple each cardiac muscle cell to the adjacent cells. It is this coupling that enables action potentials generated in one part of the heart to trigger contraction throughout the organ. Although these action potentials last up to 20 times longer than those of skeletal muscle fibers, a long refractory period prevents summation and tetanus.

**Smooth muscle** in vertebrates is found in the walls of hollow organs, such as vessels and tracts of the circulatory, digestive, and reproductive systems. In these locations it controls blood flow in the arteries, moves food through the digestive system, carries out uterine contractions during labor, and assists in regulating the temperature of the testicles. Smooth muscle is also found in the eye, where its action controls focusing and pupil diameter.

Smooth muscle cells lack striations because their actin and myosin filaments are not regularly arrayed along the length of the cell. Instead, the thick filaments are scattered throughout the cytoplasm, and the thin filaments are attached to